Junyi(Kyle) Shu

404 Westwood Plaza, Engineering VI, Room 496 – Los Angeles, CA 90095

kyleshu.github.io shujunyi@gmail.com kyleshu (1)-213-264-4111

EMPLOYMENT

University of California at Los Angeles **Los Angeles, CA, USA**

Postdoctoral Researcher *Oct 2025 – Ongoing*

- Working with Prof. Harry Xu
- Building efficient data management systems for large models and agentic AI.

Amazon Web Services **Seattle, WA, USA**

Software Development Engineer *Feb 2014 – Aug 2015*

- Built a high-performance, scalable, transactional distributed database with fellow team members.

EDUCATION

Peking University **Beijing, China**

Ph.D. in Computer Software and Theory *Sep 2021 - Jun 2025*

- Advised by Prof. Xin Jin and Prof. Xuanzhe Liu
- Dissertation: Performance Optimization towards Disaggregated Cloud Storage Systems (Outstanding Dissertation Award of PKU CS)

Peking University **Beijing, China**

Master of Engineering Management *Sep 2019 - Jul 2021*

- Advised by Prof. Xiangqun Chen
- Received the Outstanding Graduation Thesis Award.

University of California at Berkeley **Berkeley, CA, USA**

B.A. in Applied Mathematics and Computer Science *Aug 2010 - Dec 2013*

- Graduated with distinction, and was on Dean's List for several times.
- Member of Upsilon Pi Epsilon (UPE), honor society for the computing and information disciplines.

PUBLICATIONS

* = co-first author, # = corresponding author

Yunshan Jia, ***Junyi Shu#***, Chiheng Lou, Fangyue Liu, Chenbo Bi, and Xin Jin. COSMOS: Taming Heterogeneous Resource Demands in Serverless Functions with Complementary Swarming and Monotonic Scale-up. *IEEE Transactions on Networking*, 2026.

Shan Yu, Yifan Qiao, Mingyuan Ma, Yangmin Li, Shuo Yang, Xinyuan Tong, Yang Wang, Zhiqiang Xie, Yuwei An, Shiyi Cao, Ke Bao, Deepak Vij, Xiaoning Ding, Yichen Wang, Qingda Lu, Zhong Wang, Gao Gao, Harry Xu#, ***Junyi Shu#***, Jiarong Xing#, and Ying Sheng#. Prism: Cost-Efficient Multi-LLM Serving via GPU Memory Ballooning. In *20th USENIX Symposium on Operating Systems Design and Implementation (OSDI 26)*, pages 75–93, Seattle, WA, July 2026. USENIX Association.

Junyi Shu, Xiaolong Huang, Gang Huang, Hong Mei, Xuanzhe Liu, and Xin Jin#. Serverless Replication of Object Storage across Multi-Vendor Clouds and Regions. In *Proceedings of the 21st European Conference on Computer Systems*, EUROSYS '26, page 1780–1796, New York, NY, USA, 2026. Association for Computing Machinery.

Junyi Shu, Kun Qian, Ennan Zhai, Xuanzhe Liu, and Xin Jin#. Burstable Cloud Block Storage with Data Processing Units. In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, pages 783–799, Santa Clara, CA, July 2024. USENIX Association.

Junyi Shu, Ruidong Zhu, Yun Ma, Gang Huang#, Hong Mei, Xuanzhe Liu, and Xin Jin#. Disaggregated RAID Storage in Modern Datacenters. In *Proceedings of the 28th ACM International Conference on Architectural*

Support for Programming Languages and Operating Systems, Volume 3, ASPLOS 2023, page 147–163, New York, NY, USA, 2023. Association for Computing Machinery.

Junyi Shu, Xin Jin, Yun Ma, Xuanzhe Liu, and Gang Huang. Cost-effective data analytics across multiple cloud regions. In *Proceedings of the SIGCOMM '21 Poster and Demo Sessions*, SIGCOMM '21, page 1–3, New York, NY, USA, 2021. Association for Computing Machinery.

SERVICES

- **Conference Program Committee**: ACM International Conference on Architectural Support for Programming Languages and Operating Systems - ASPLOS (2026); The IEEE/ACM International Symposium on Microarchitecture - MICRO (2026)
- **Journal Reviewer**: IEEE Transactions on Parallel and Distributed Systems (2026); ACM Transactions on Storage (2025); ACM Transactions on Architecture and Code Optimization (2025, 2026)
- **Shadow Program Committee**: ACM European Conference on Computer Systems - EuroSys (2026)

INTERNSHIP

Alibaba Cloud **Beijing, China**
Research Intern *May 2023 – Jul 2024*
- Designed and developed a software module that supports burstable I/Os while limiting tenant interference.

Amazon Instant Video **Seattle, WA, USA**
Software Development Engineer Intern *Jun 2013 – Aug 2013*
- Explored available metrics for evaluation video quality and developed a library to extract them.